

Beyond Recall: Operational Correctness for Agent Memory Systems

OpMem v0 Benchmark and Empirical Pins

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Abstract

Retrieval benchmarks such as LoCoMo and LongMemEval measure whether an agent recalls the right fact, but not whether a memory *system* honors forget, correct, isolate, and supersede semantics in production. We present **OpMem v0**, a 13-task, pass/fail suite over neutral **remember/recall/revise/forget** scripts with substring-based assertions on ranked results. On the frozen 2026-08-15 pin (git SHA `1b5ab3e`), Brainy passes **13/13** via `GET /memories/search` while Mem0 **Platform** passes **10/13** (failures: `cor02`, `sup03`, `upd02`). This manuscript reports repository evidence, a reproducible protocol, threats to validity, and new embedded-harness re-measurements on the current merge-pin git revision where completed. We do *not* claim conversational-recall superiority over Mem0; OpMem operational leadership is orthogonal to LoCoMo J-scores.

1 Introduction

Production agents fail when suppressed memories leak after re-ingestion, corrections do not stick, tenants bleed across isolation boundaries, or stale facts outrank updates. Conversational QA benchmarks [3, 5] do not encode these lifecycle contracts; hallucination-oriented memory benchmarks such as HaluMem [2] focus on factual faithfulness rather than governed delete/correct semantics.

OpMem targets **operational correctness**: deterministic scripts, binary scoring, and explicit adapter mappings to each system’s native API surface [1]. Section 8 lists threats; Section 6 separates **completed** evidence from **blocked** external comparisons.

2 Related work

Retrieval QA. LoCoMo [3] and LongMemEval [5] score end-to-end answer quality over long conversations; they are not substitutes for lifecycle tests.

Memory products. Mem0 documents accumulation-oriented designs and names staleness as an open problem [4]. We report Mem0 **Platform** numbers only on OpMem fixtures; we do not import Mem0 blog LoCoMo percentages as same-pin scores.

Pending external comparisons. Paid or vendor-hosted counter-runs are allowed only after a written cost estimate and explicit approval; that gate was not a categorical denial of spend. Mem0 Platform on OpMem v0 is **completed** at negligible API volume (13 tasks, 47 scripted steps; see Section 6.8). Mem0 OSS, LangMem, and Letta (pip server) have **completed** Phase 2 live pins (Section 6.8); Zep remains **blocked** on missing `ZEP_API_KEY` injection in the cloud agent VM (verified 2026-09-23; no Zep score claimed).

3 Neutral operation model

Actors are (tenant, subject) pairs. Tasks are operation scripts; `revise` and `forget` reference prior `remember` step ids. Adapters namespace tenants per task (`opmem-{nonce}-{task}`) for hermetic runs.

Table 1: Adapter mappings (normative; from `PUBLICATION_READINESS.md`).

Op	Brainy	Verbatim baseline	Mem0 Platform
remember	POST /ingest	append	POST /v1/memories/
recall	GET /memories/search	token overlap	POST /v2/memories/search/
revise	POST .../correct	in-place replace	PUT /v1/memories/{id}/
forget	POST .../suppress	delete	DELETE /v1/memories/{id}/

4 Task suite

OpMem v0 comprises 13 JSON fixtures under `fixtures/opmem/`: three suppression, three correction, three isolation, three staleness (including `upd03_state_supersession`), and one idempotency task. Assertions use case-insensitive substring checks on ranked `content` fields (`min_results`, `top_contains`, `must_exclude`, etc.).

5 Methodology

5.1 Scoring lane

Primary pins use Brainy `/memories/search`, not product `POST /recall` (Section 6.4 reports a `/recall` lane ablation). The harness (`evals/run_opmem.py`) exits non-zero only on infrastructure errors, not task failures—CI proves runner health, not 13/13, unless JSON is parsed.

5.2 Reproduction commands

```
go test ./internal/api/ -run TestOpMemBenchmarkAgainstHTTPServer -count=1
python3 evals/run_opmem.py --systems verbatim,brainy \
  --base-url http://127.0.0.1:8080 --json-out opmem-report.json
```

Paper experiments: `scripts/run-opmem-manuscript.sh` and `evals/run_opmem_paper_experiments.py`.

6 Results

6.1 Frozen 2026-08-15 pin (historical, SHA 1b5ab3e)

[completed] Brainy **13/13**; verbatim **10/13**; Mem0 Platform **10/13** (artifacts `opmem-fresh-local-20260815.m`
`opmem-mem0-fresh-20260815.md`).

Mem0 failures on that pin: `cor02` (correction stickiness), `sup03` (durable forget leak), `upd02` (preference ranking).

Table 2: Category breakdown on 2026-08-15 Mem0 Platform counter-run.

Category	Mem0 Platform	Brainy (same cycle)
correction	2/3	3/3
isolation	3/3	3/3
suppression	2/3	3/3
staleness	2/3	3/3
idempotency	1/1	1/1
overall	10/13	13/13

6.2 Current-SHA re-measurement

[**completed**] At git SHA 20533f2ca83e (UTC 2026-09-18T23:12:29Z), embedded API harness: Brainy (search lane) **13/13**, verbatim **10/13**. Mem0 Platform (same run): **10/13** (API surface: Platform via Mem0OpAdapter).

6.3 Combined machine-readable artifact

[**completed**] Committed `opmem-primary.json` lists per-task outcomes for verbatim, brainy, mem0 with manifest fields (SHA, date, fixture count).

6.4 Recall vs search lane

Table 3: Brainy recall lane and revise mapping (SHA 20533f2ca83e).

Configuration	Overall
brainy	13/13
brainy-recall	9/13
brainy-supersede	11/13

Search (`/memories/search`) is the normative pin lane. `brainy-recall` uses `POST /recall` (context mode). Correction tasks on the recall lane may return HTTP 409 during `/correct` (infrastructure errors in artifact JSON).

6.5 Correct vs supersede mapping

[**completed**] Mapping `revise` $\rightarrow$ `POST .../supersede` yields **11/13** vs **13/13** with `/correct` on the search lane (same SHA). Failures are task-level assertion misses, not missing runs.

6.6 Mechanism breakdown

Category pass rates separate correction/suppression/staleness contracts; verbatim approximates rank-only recall without governed lifecycle ops.

6.7 Statistical stability

[**completed**] Five consecutive Brainy search-lane repeats on one embedded server session. **Valid runs only** (infra errors = 0): 2/5; scores: 13/13.

Category	Brainy	Verbatim (rank-only)
correction	3/3	2/3
idempotency	1/1	0/1
isolation	3/3	3/3
staleness	3/3	3/3
suppression	3/3	2/3

- Repeat 1: 13/13 (infra errors=0)
- Repeat 2: 9/10 (infra errors=3)
- Repeat 3: 9/10 (infra errors=3)
- Repeat 4: 13/13 (infra errors=0)
- Repeat 5: 9/10 (infra errors=3)

Repeats with infra errors > 0 (3 of 5) are **invalid/incomplete**: the harness omits errored tasks from the denominator, so aggregates such as 9/10 are *not* comparable to a clean 13/13 pin. Use isolated `TestOpMemBenchmarkAgainstHTTPServer` for merge-gate stability.

6.8 External systems

[**completed**] Mem0 **Platform**: 10/13 on SHA 20533f2 (`opmem-pin-20260918.json`); same headline as 2026-08-15 pin. Completed under negligible Platform API use (full suite ≈ 47 scripted steps).

[**completed**] Mem0 **OSS** (local Qdrant path, `mem0ai 2.1.0`): **9/13** on run `opmem-phase2-full-mem0` (13 tasks, 47 steps; 0 infra errors). Failures: `sup03`, `upd01--upd03`. Artifact: `live-full-opmem-phase2-full-mem0`

[**completed**] LangMem (Postgres 16 + `pgvector 0.6`, `langmem 0.0.30`): **8/13** on run `opmem-phase2-full1` (0 infra errors). Failures: `cor02`, `iso03`, `sup01--sup03`. Artifact: `live-full-opmem-phase2-full1.json`.

[**completed**] Letta (pip server `letta==0.11.7`, client `0.1.307`, SQLite): **5/13** on run `opmem-phase2-letta-full` (0 infra errors). Adapter: per-task agent + `opmem_facts` block; lexical recall. Failures: `cor01--cor03`, `sup01--sup03`, `upd01--upd02`. Artifact: `live-full-opmem-phase2-letta-full.json`. [**blocked**]

Zep Cloud (\$0 paid): on 2026-09-23 VM check, `ZEP_API_KEY` absent from env and from `CLOUD_AGENT_INJECTED_SECRET` (20 names listed in `zep-env-check-2026-09-23.json`); `opmem-phase2-zep-full` not executed.

[**completed**] Payload budget worksheet (tiktoken on serialized requests incl. mem0 extraction prompt): worst-case OpenAI **\$0.15** total vs \$3 ceiling (`cost-worksheet-payload-budget.json`). Legacy dry-run mock meter (\$0.0008) is not used for spend approval.

6.9 Cost-estimation plan (before any new paid run)

OpMem v0 has 13 tasks and 47 scripted steps (`fixtures/opmem/`). For each pending system, prepare billing before requesting approval:

- **Phase 1–2** — `evals/opmem_external/`: deterministic namespaces, per-task reset, payload budget (`cost-worksheet-payload-budget.json`), live harness `run_opmem_external_live.py`. Setup: `external-lanes-setup.md`.
- **Mem0 OSS** — Local Qdrant + `OPENAI_API_KEY`; ceiling \$1; dry-run projected OpenAI spend in worksheet.

- **Zep** — Zep Cloud free tier only (\$0 paid); cannot pin OpenAI models (product constraint recorded).
- **Letta** — Self-hosted App Server; ceiling \$1; no legacy Docker API or Letta Cloud paid plan.
- **LangMem** — Postgres/pgvector; ceiling \$1; no `InMemoryStore` on measured runs.

No dollar totals are asserted here until rate cards and deployment mode are pinned; approval requires the completed worksheet, not a categorical spend ban.

7 Discussion

OpMem failures on Mem0 align with accumulation-oriented update semantics: corrections and durable forgets are not first-class lifecycle events. Verbatim baseline passes 10/13, showing the suite discriminates rank-only stores from governed suppression/correction. Policy disagreements remain (sup03 durable forget is a normative contract).

8 Limitations

1. Small $n = 13$; each task is $\sim 7.7\%$ of the aggregate score.
2. Substring scoring does not guarantee semantic safety.
3. Search vs `/recall` product path may diverge (lane ablation quantifies on Brainy).
4. English, domain-neutral fixtures; no concurrency or TTL tests.
5. Mem0 Platform API drift; Zep/Letta not measured on Phase 2 host (credential/server block). OSS/LangMem pins are single-pass, small n .
6. Fixture authorship bias; no independent external audit documented.

9 Artifact appendix

Committed run manifests and JSON under `docs/benchmarks/artifacts/opmem-manuscript-*:run-manifest.json`, `opmem-primary.json`, `opmem-lane-ablation.json`, `opmem-stability.json`, `opmem-mechanism.json`, and `commands.log`.

Build PDF: `make -C docs/research/opmem/paper pdf`.

10 Future work

v1 task expansion (~ 30 tasks), Zep/Letta/LangMem adapters, supersede-native revise mapping as default where lineage is required, and fail-closed CI assertion of 13/13 on merge pins.

References

- [1] Brainy. OpMem v0 specification and harness. <https://github.com/tryvinci/brainy/docs/research/opmem-spec.md>, 2026.

- [2] Ding Chen, Simin Niu, Kehang Li, Peng Liu, Xiangping Zheng, Bo Tang, Xinchu Li, Feiyu Xiong, and Zhiyu Li. HaluMem: Evaluating hallucinations in memory systems of agents. *arXiv preprint arXiv:2511.03506*, 2025.
- [3] Adyasha Maharana et al. LoCoMo: Evaluating very long-term conversational memory of llm agents. *arXiv preprint arXiv:2402.17710*, 2024.
- [4] Mem0. State of AI agent memory (2026). <https://mem0.ai/blog/state-of-ai-agent-memory-2026>, 2026. Vendor context on staleness and cross-session evolution; not an OpMem same-pin score.
- [5] Howard Yen Wu et al. LongMemEval: Benchmarking long-term memory in llm chat assistants. *arXiv preprint arXiv:2410.10819*, 2024.